

BioMaster Druggable Proteome Screening Report

Computational Drug-Target Prioritization Workflow

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Executive Summary

This report summarizes a computational screen of FDA-approved small molecules against the ChEMBL druggable proteome. The workflow first predicts drug-protein affinity with ConPlex, then performs structure-level prioritization with AlphaFold receptor preparation and DiffDock docking for high-priority representatives.

The outputs are intended for candidate prioritization and expert review. They are not claims of clinical efficacy or experimentally confirmed binding.

Metric	Value
Usable FDA-approved small molecules	915
Valid ChEMBL druggable protein records	5306
Unique protein sequences	891
Full ConPlex affinity matrix	4854990
Top affinity candidates	10000
DiffDock representative complexes	1872
Completed DiffDock representatives	1370 (73.2%)
Missing or failed DiffDock representatives	502 (26.8%)
Top10000 rows with mapped completed structure	7216 (72.2%)

Workflow

1. Standardize the FDA-approved small-molecule library and ChEMBL druggable protein table.
2. Collapse identical protein sequences to avoid redundant ConPlex inference.
3. Predict affinity for 815,265 unique drug-sequence pairs and expand scores back to 4,854,990 drug-target pairs.
4. Select the Top10000 affinity candidates.
5. Collapse Top10000 candidates by drug and protein sequence, yielding 1,872 representative docking jobs.
6. Prepare AlphaFold receptor structures and run DiffDock to generate rank-1 ligand poses and confidence scores.
7. Merge DiffDock results back to both the representative table and the full Top10000 candidate table.

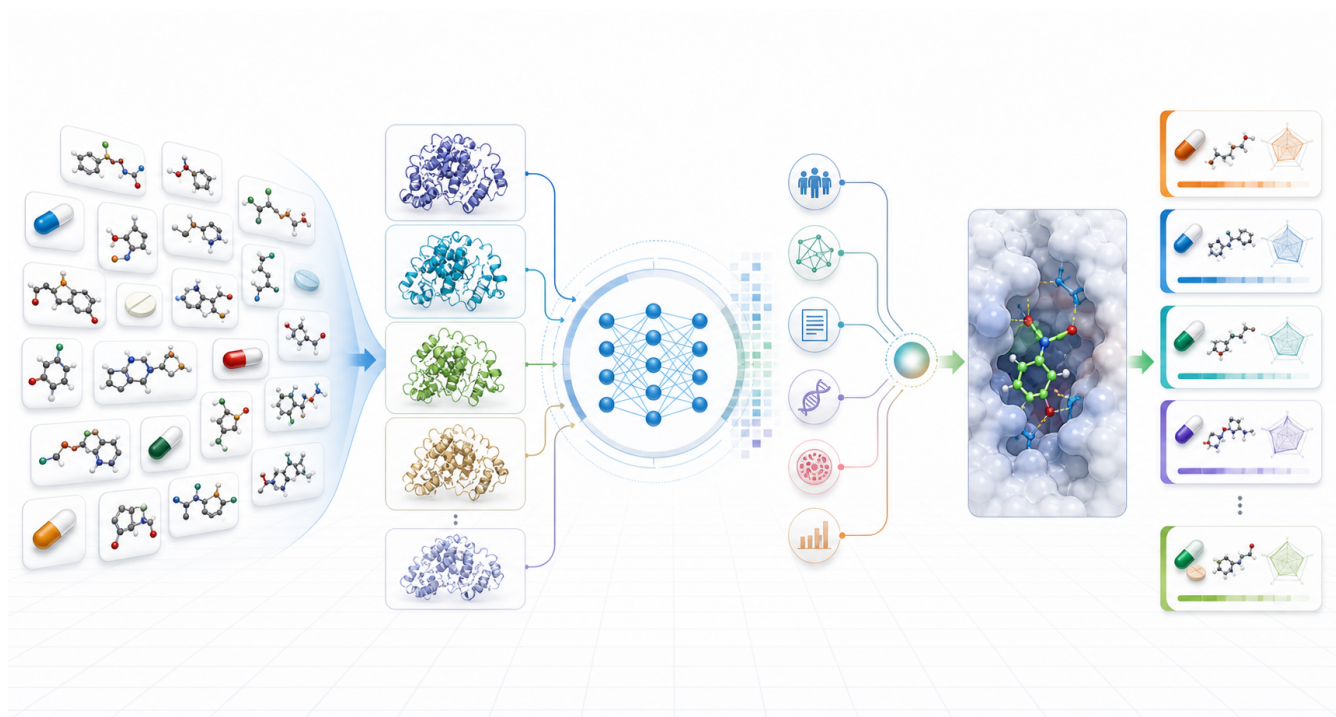


Figure 1: BioMaster screening concept: small-molecule library, druggable protein targets, AI-DTI affinity scoring, structure-level docking, and prioritized candidate outputs.

Model Roles

ConPlex is used for broad drug-target affinity prioritization from chemical and protein-sequence representations. Its score is a ranking signal, not a measured binding constant.

AlphaFold provides receptor structures for selected target representatives. Long receptors are cropped only when required for DiffDock feasibility.

DiffDock generates rank-1 ligand poses and confidence scores for high-priority representatives. The confidence score is used as structure-level supporting evidence.

Representative Results

Rank	Drug	Gene	Protein	Affinity	Docking	Consensus	Status
1	Doxazosin Mesylate	ADRA1A	P35348	0.97390723	NA	0.992401000	missing_output
2	Degarelix Acetate	PTAFR	P25105	0.9689111	NA	0.987310000	missing_output
3	Degarelix Acetate	GNRHR	P30968	0.9656983	NA	0.984036000	missing_output
4	Afatinib Dimaleate	EGFR	P00533	0.96460074	NA	0.982917000	missing_output
5	Buprenorphine	OPRK1	P41145	0.9595081	NA	0.977728000	missing_output
6	Atropine	CHRM2	P08172	0.95824254	NA	0.976438000	missing_output
7	Degarelix Acetate	OXTR	P30559	0.9576171	NA	0.975801000	missing_output
8	Degarelix Acetate	P2RY2	P41231	0.9541216	NA	0.972239000	missing_output
9	Naltrexone	OPRK1	P41145	0.95833254	-0.45	0.969219266	completed
10	Atropine	CHRM4	P08173	0.9499203	NA	0.967958000	missing_output
11	Degarelix Acetate	P2RY12	Q9H244	0.9496888	NA	0.967722000	missing_output
12	Atropine	HTR1A	P08908	0.9489648	NA	0.966984000	missing_output
13	Atropine	CHRM1	P11229	0.9483994	NA	0.966408000	missing_output
14	Degarelix Acetate	TACR2	P21452	0.9481127	NA	0.966116000	missing_output
15	Atropine	DRD3	P35462	0.9480593	NA	0.966062000	missing_output

Rank	Drug	Gene	Protein	Affinity	Docking	Consensus	Status
16	Dorzolamide Hydrochloride	CA2	P00918	0.9533263	-0.36	0.966016917	completed
17	Degarelix Acetate	AVPR1A	P37288	0.94641465	NA	0.964386000	missing_output
18	Degarelix Acetate	GHRHR	Q02643	0.9452929	NA	0.963243000	missing_output

Docking Audit

Status	Count	Receptor preparation status	Count
Completed	1370	full_length_ok	1833
Missing output	502	curated_auto_high_confidence_crop	34
		curated_EGFR_kinase-domain_crop	4
		curated_JAK1_kinase-domain_crop	1

Interpretation

Open Targets is not required for the current primary screen. It is a target-disease association resource and should be reintroduced when the analysis is focused on a specific disease, cancer type, mutation, or molecular subtype. In the current druggable-proteome workflow, the first objective is to identify high-affinity drug-target candidates within a pharmacologically relevant target space.

Recommended Next Step

The next practical step is to review the structure-enhanced Top100 candidates, remove pairs with weak biological rationale or problematic docking output, and select 20–50 candidates for deeper expert review. A smaller 5–20 candidate subset can then be chosen for literature review, orthogonal docking, or experimental validation.